

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: Power BI And Tableau Practical

PRACTICAL - 1

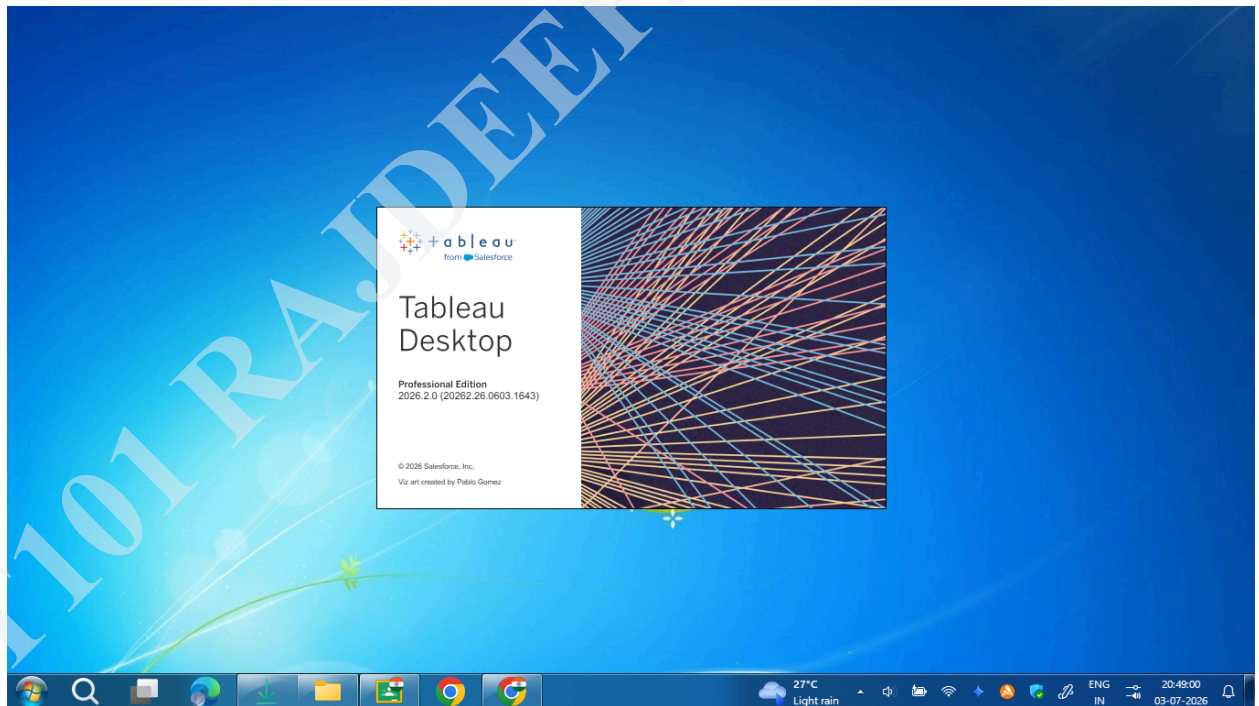
AIM: Getting Started with Tableau and Building Basic Visualisations.

- a) Connect to a CSV dataset and explore the Tableau interface
- b) Identify and differentiate Measures and Dimensions
- c) Convert fields between discrete and continuous and observe changes
- d) Create and customise a basic bar chart
- e) Build a line chart for time-series analysis
- f) Create a geographic map visualisation
- g) Combine multiple charts into a simple interactive dashboard

Steps:-

Part A: Connect to the Dataset & Explore the Interface

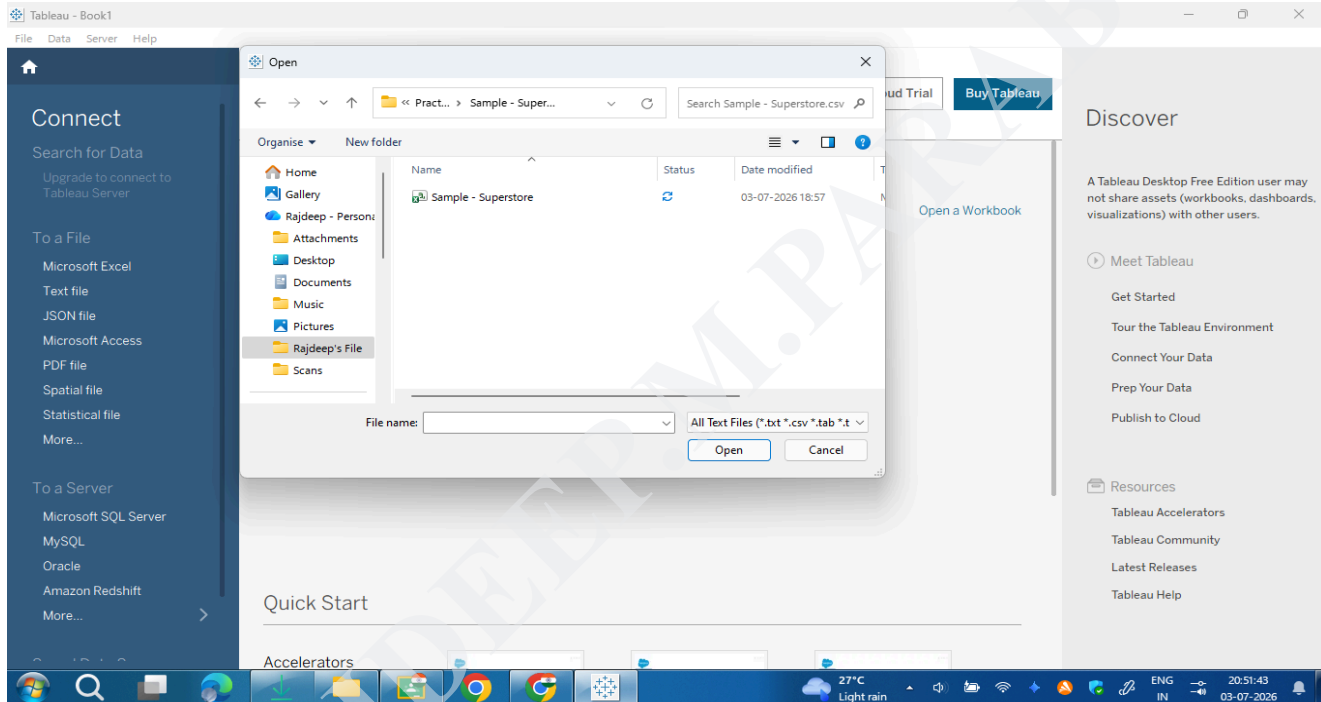
1. Open Tableau: Launch Tableau Desktop on your Windows computer and Sign In.



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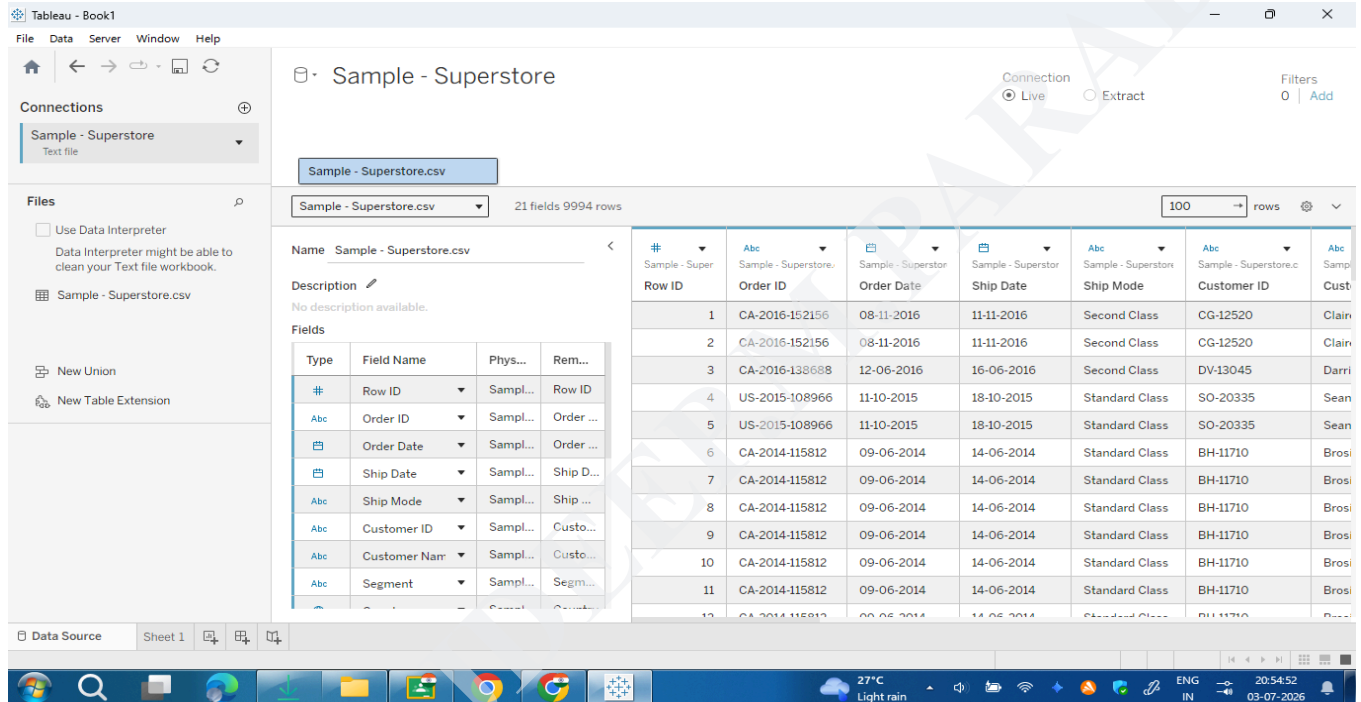
2. Select File Type: On the blue panel on the left side under Connect, click on Text File (Tableau processes CSV files as text files).
3. Upload Dataset: A Windows File Explorer window will pop up. Navigate to where your downloaded Superstore CSV file is saved, select it, and click Open.



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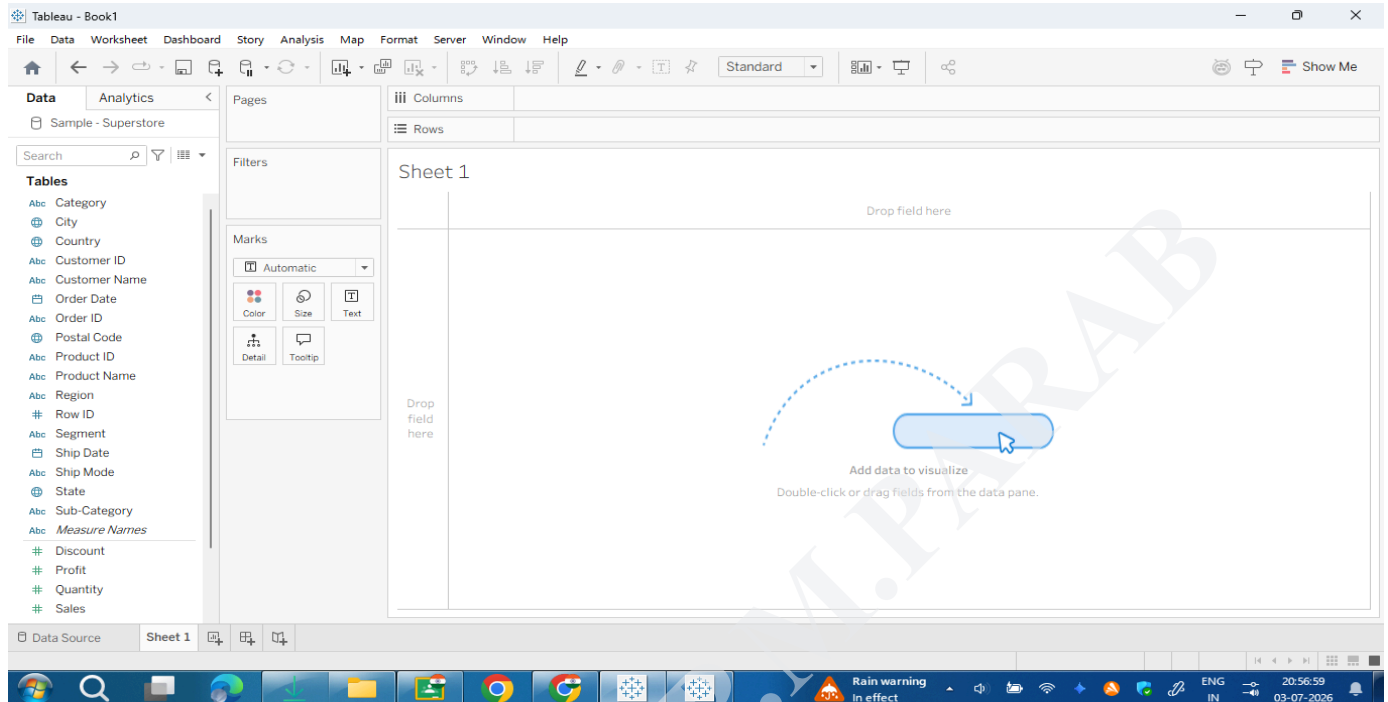
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- Go to Worksheet: You will see the Data Source page. Look at the bottom to preview your data rows to ensure they look right. Then, click the Sheet 1 tab at the bottom-left corner to enter your workspace. Tableau will automatically set up a native workspace and open your first sheet.



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Interface Navigation Guide

- Data Pane (Far Left): This area displays all the variables and columns from your dataset.
- Shelves (Top/Centre): The Columns and Rows shelves act as your X and Y grid coordinates to position your data.
- Marks Card (Left of Centre): This panel controls visual properties like colour, text labels, and sizing.

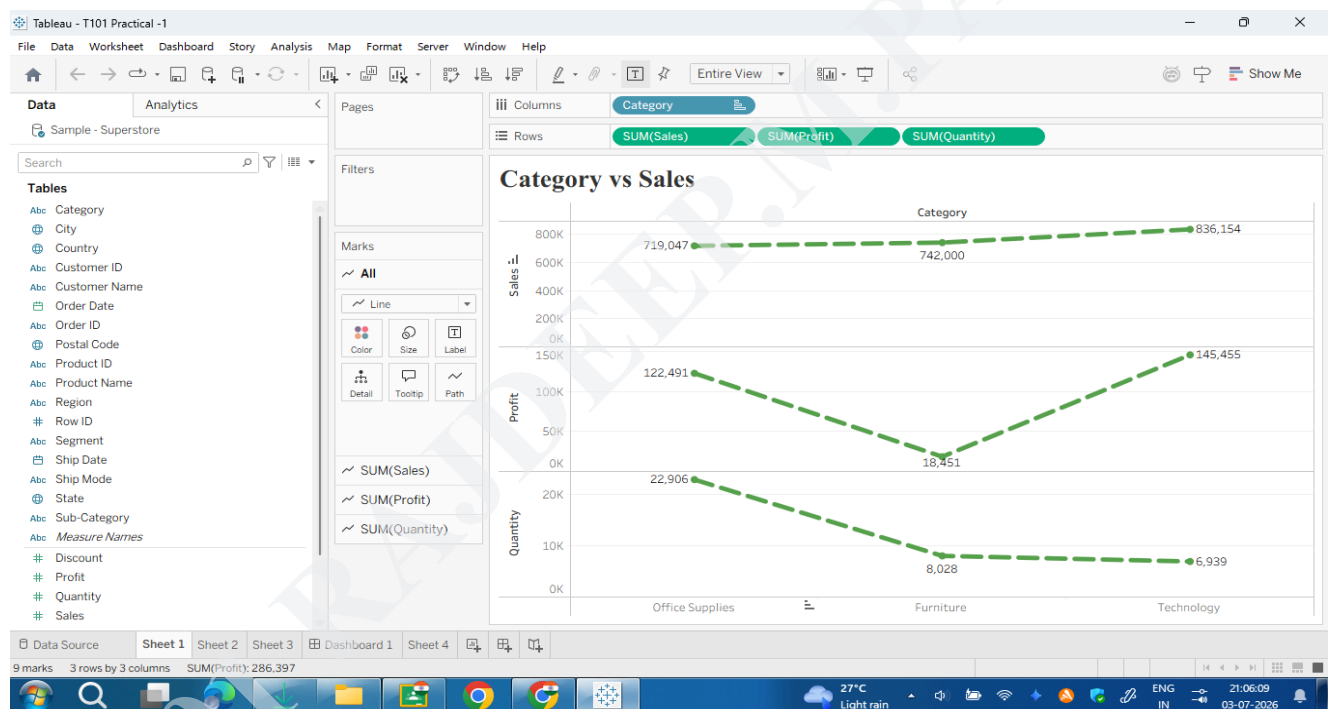
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Part B: Identify and Differentiate Measures and Dimensions

Look at the Data Pane on the left. Tableau reads your file data and automatically separates it into two categories:

- Dimensions (Blue Icons): These are text, date, or geographic fields (e.g., *Category*, *Segment*, *State*, *Order Date*). They are used to slice, bucket, and group your data.
- Measures (Green Icons): These are raw numerical values (e.g., *Sales*, *Profit*, *Quantity*). Tableau automatically calculates sums or averages for these numbers when you drop them onto a chart.



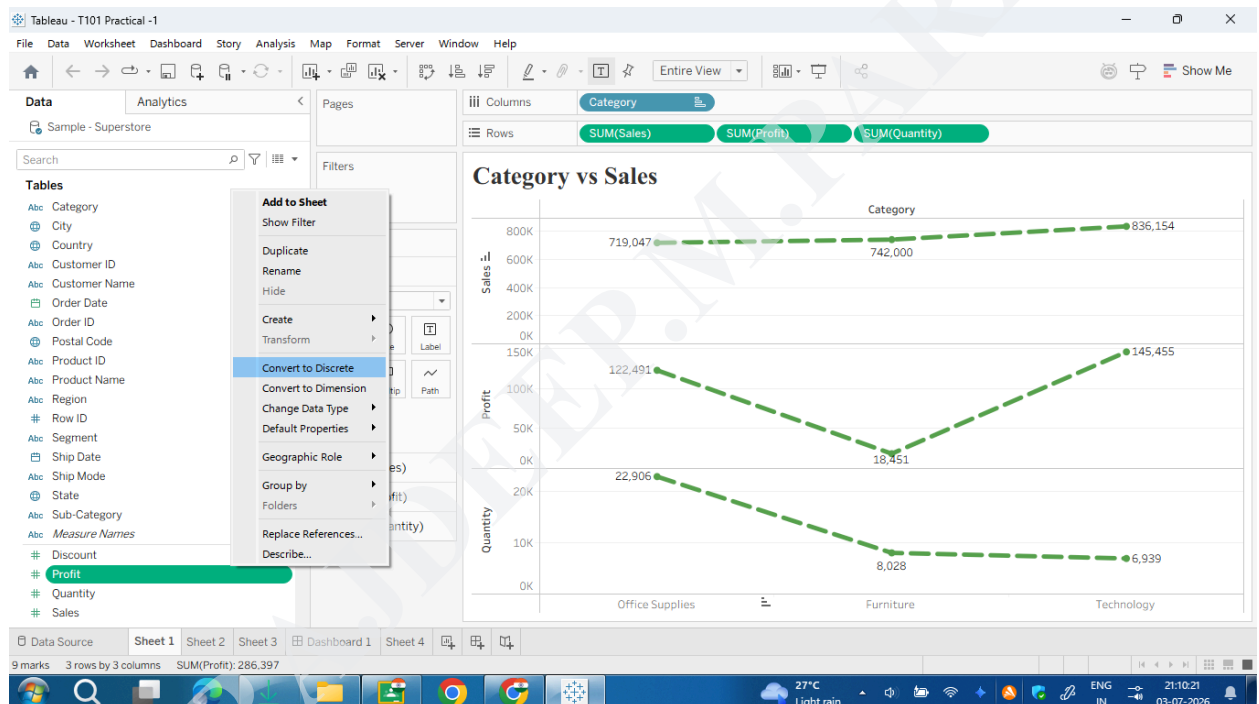
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Part C: Convert Fields Between Discrete and Continuous

Tableau uses colours to show how it treats data. Blue means Discrete (creates separate, distinct header blocks) and Green means Continuous (creates an unbroken, smooth numeric axis line).

1. Find the Quantity field in your green measures list on the left.
2. Right-click on Quantity and select Convert to Discrete. Notice its icon changes from a green hash symbol (#) to a blue hash symbol (#).

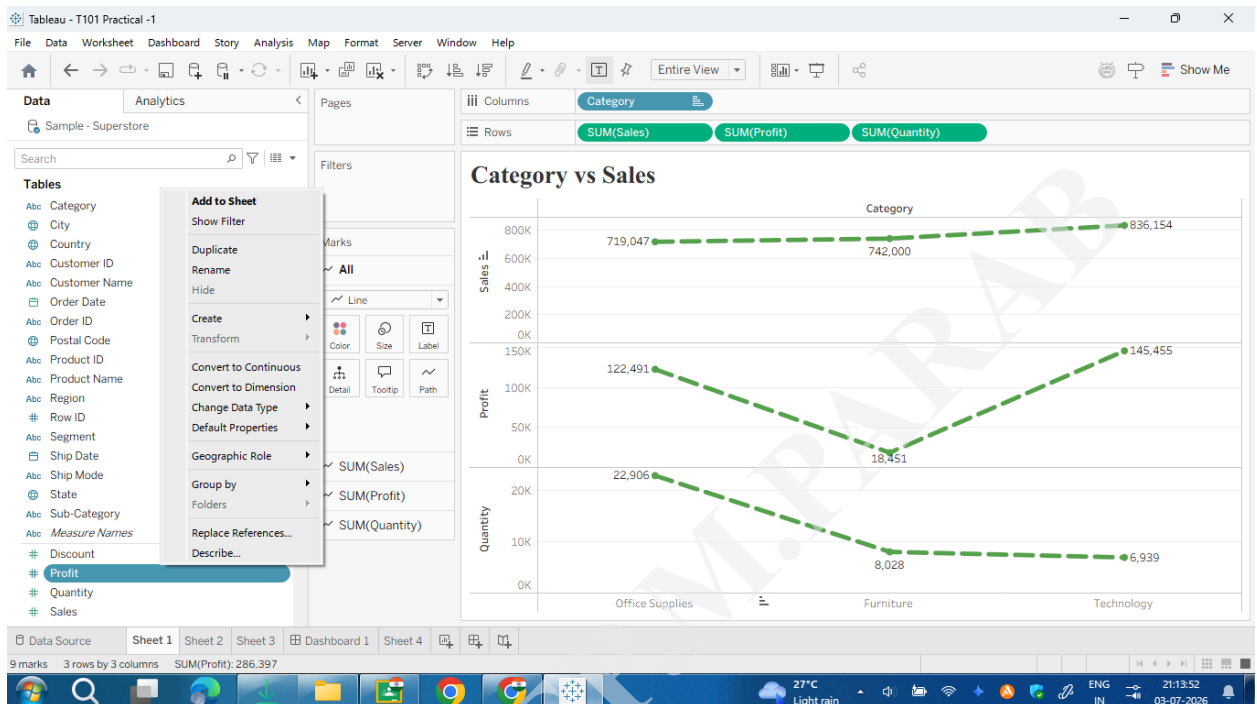


3. Drag this new blue Quantity pill and drop it onto the Columns shelf. It will display separate, isolated header columns for each exact number (1, 2, 3, etc.).
4. Drag the Quantity pill back off the shelf to remove it.

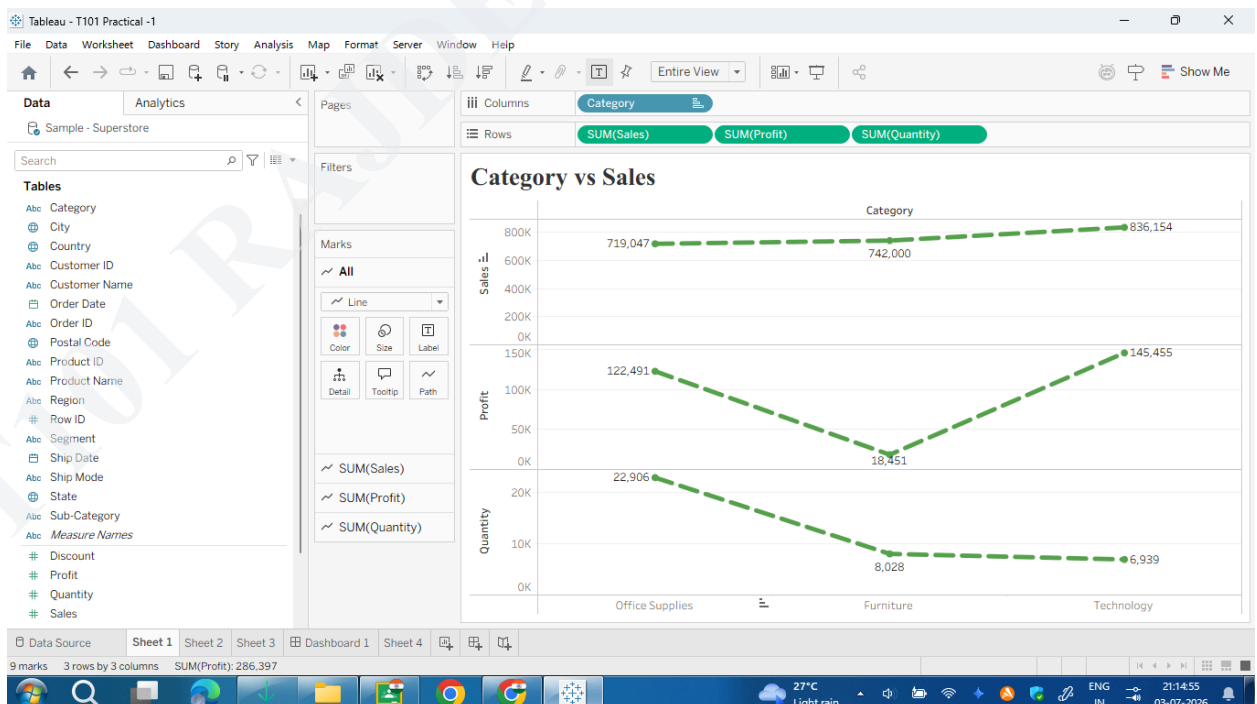
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5. Right-click Quantity in the Data pane again and select Convert to Continuous.



6. Drag it back onto the Columns shelf. Instead of broken-up column blocks or lines, Tableau now draws a single, smooth, continuous numeric axis line.

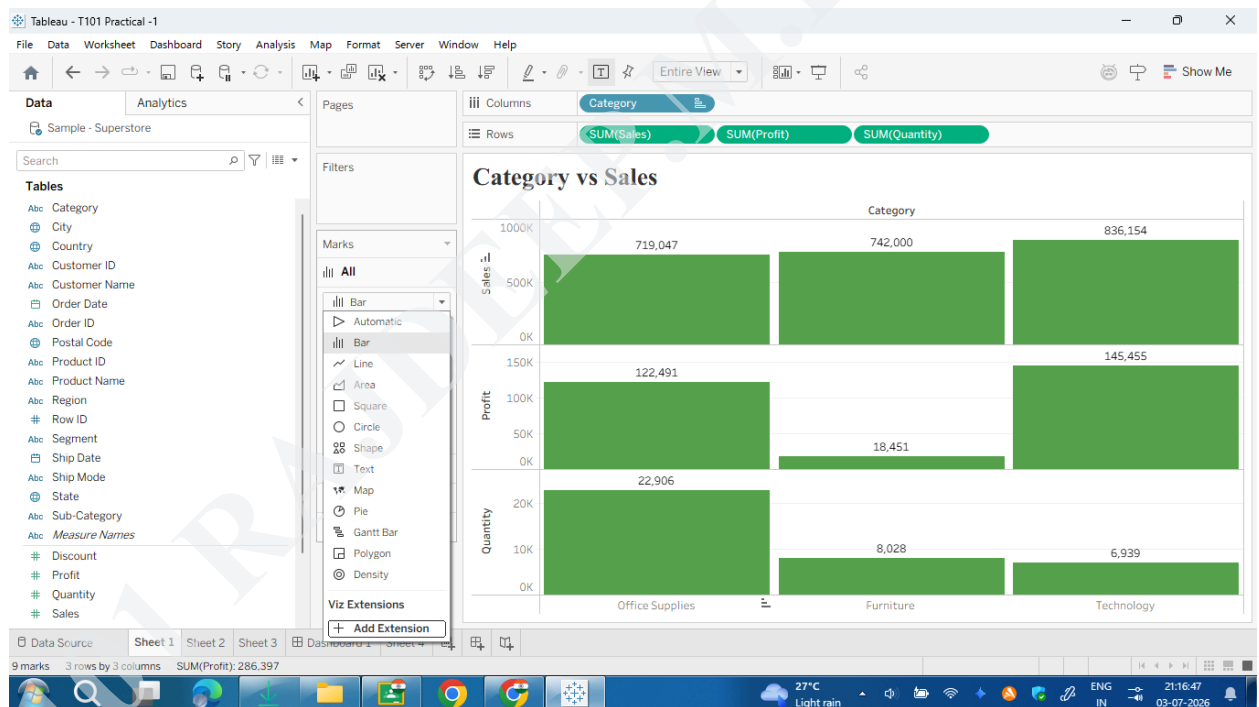


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Part D: Create and Customise a Basic Bar Chart

1. In the Data Pane, find the **Category** dimension and drag it onto the **Columns** shelf.
2. Find the **Sales** measure and drag it onto the **Rows** shelf. You will instantly see three vertical bars for *Furniture*, *Office Supplies*, and *Technology*.
3. **Add Colour:** Grab the **Category** dimension from the Data pane a second time, and drop it directly onto the **Colour** tile inside the **Marks** card.
4. **Add Data Labels:** Drag the **Sales** measure from the Data pane and drop it directly onto the **Label** tile inside the **Marks** card to display exact revenue numbers above the bars.
5. **Rename Sheet:** Double-click the sheet tab named **Sheet 1** at the very bottom, type **Category Bar Chart**, and press Enter.

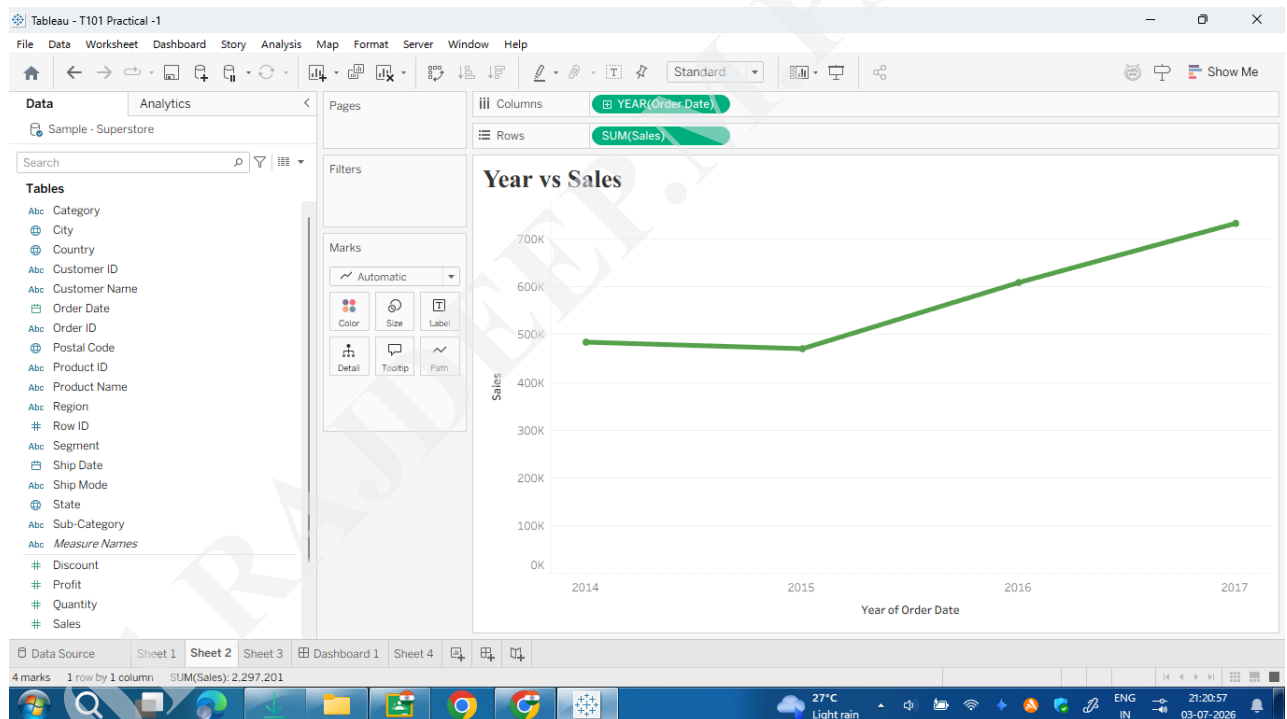


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Part E: Build a Line Chart for Time-Series Analysis

1. Click the **New Worksheet** button on the bottom tab bar (the small icon showing a chart with a plus sign).
2. Drag the **Order Date** dimension onto the **Columns** shelf.
3. Drag the **Sales** measure onto the **Rows** shelf. Because you put a date field on the columns shelf, Tableau automatically generates a line chart.
4. **Drill Down into Dates:** Look at the blue **YEAR(Order Date)** pill sitting on your Columns shelf. Click the small plus symbol (+) on that blue pill. It will expand your timeline into Quarters and Months, revealing deeper sales trends over time.
5. Double-click the sheet tab at the bottom, rename it to **Sales Trend**, and press Enter.

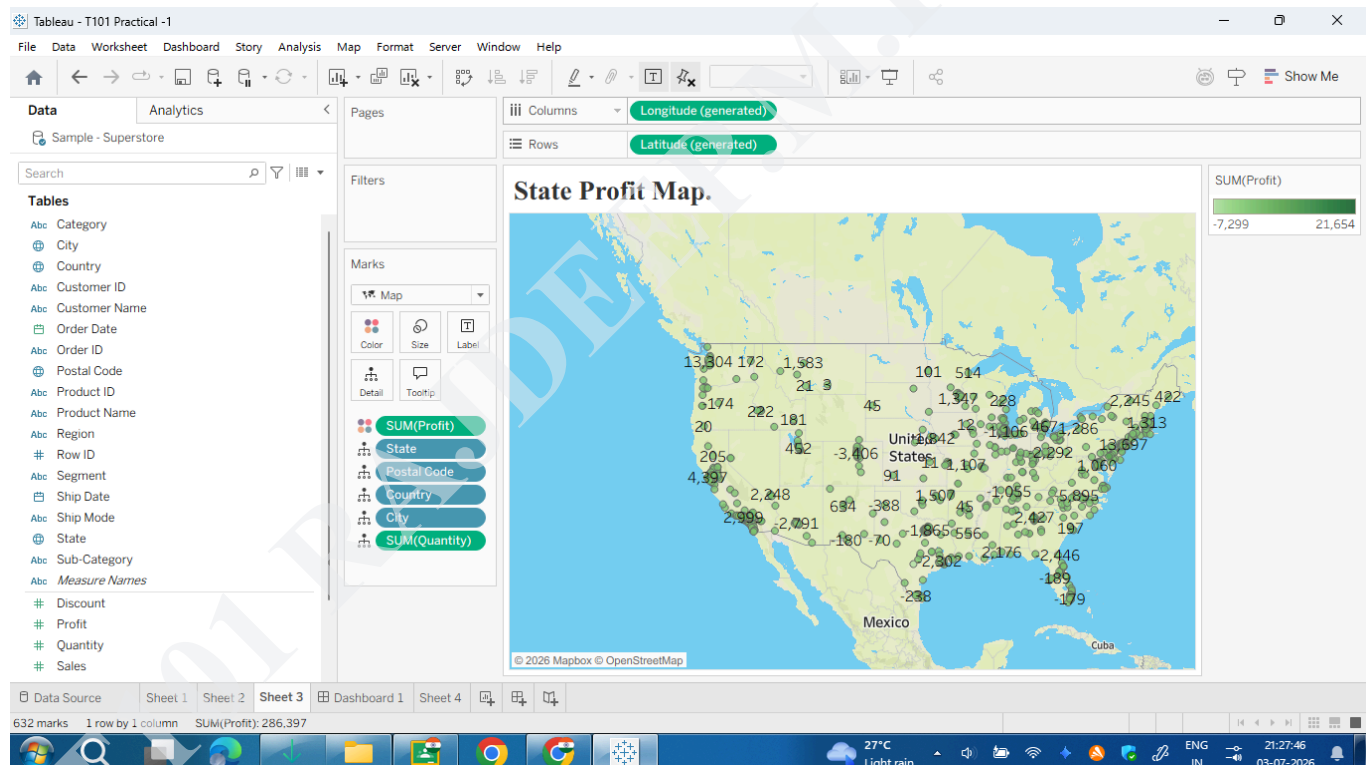


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Part F: Create a Geographic Map Visualisation

1. Click the **New Worksheet** button at the bottom to open a third blank sheet.
2. In the Data Pane, find the **State** field. It will feature a tiny globe icon next to it, showing that it contains geographic data.
3. **Double-click** the **State** field. Tableau will automatically input Latitude and Longitude coordinates onto your shelves and generate a map.
4. **Colour-Code Map:** Go to your Data Pane, pick up **Profit**, and drop it directly onto the **Colour** tile inside the **Marks** card. These colours are based on each state's financial performance.
5. Double-click the sheet tab at the bottom, rename it to **State Profit Map**, and press Enter.

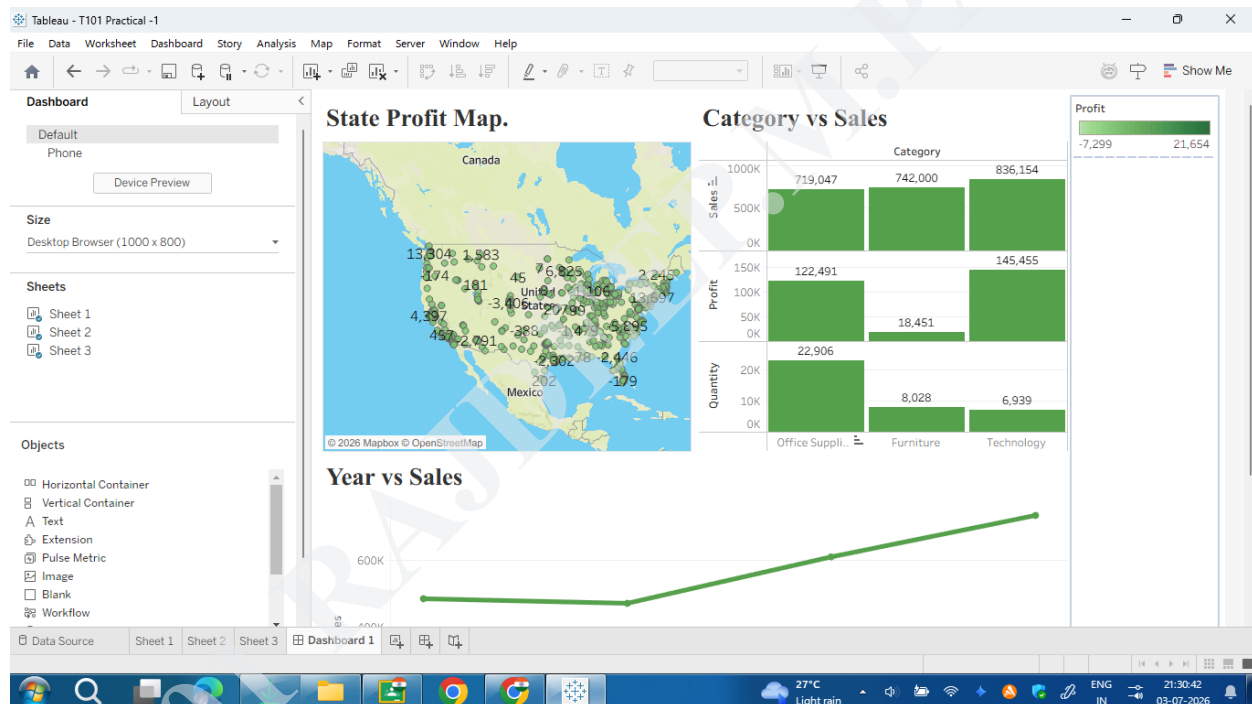


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Part G: Combine Charts into an Interactive Dashboard

1. Look at the bottom navigation bar and click the **New Dashboard** icon (it looks like a small 2x2 grid window pane with a plus sign).
2. On the left-side panel, you will see your three sheets listed: *Category Bar Chart*, *Sales Trend*, and *State Profit Map*.
3. Drag the **Category Bar Chart** sheet and drop it onto the empty dashboard canvas area.
4. Drag the **Sales Trend** sheet and drop it right along the bottom edge of the bar chart space.
5. Drag the **State Profit Map** sheet and drop it along the right side of the layout.



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